

▣ 25 χ^2 tests of independence

Example 1

A medical study is comparing the 10-year survival rate for 190 heart transplant patients in a number of countries.

The data is recorded in a [CSV](#) file and loaded in to R as a data.frame called *heart*. The output from a call to *head()* is shown opposite.

```
> head(heart)
  country survived
1      US      yes
2      UK      yes
3      NZ      yes
4      US     no
5      US     no
6      US     no
```

A useful question to consider is whether there are difference in survival rates between the countries.

Solution

Here we have two categorical variables – country is nominal (3 levels) and survived in binary. Thus, the t-test or ANOVA (which require that one variable be continuous) are of no use – we need something else.

First, we construct a cross-tabulation in R (left). The neatly formatted output is shown on the right.

```
> table(survived, country)
      country
survived NZ UK US
no       5 28 29
yes     32 34 62
```

Country	New Zealand	UK	US
Did not survive	5	28	29
Survived	32	34	62

We can visualised this data in R with `> plot(table(country, survived))`

The output is shown opposite and suggests that there may be differences between the countries as the yes:no ratio appears to vary from very high in New Zealand down to about 1:1 in the UK.

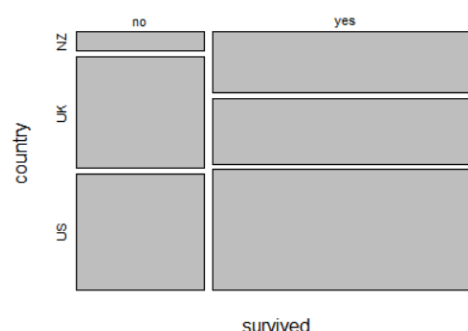
But of course, we would like to establish if the differences are beyond what might have been expected due to sampling error – i.e. is there a statistically significant difference.



Notes

If we swapped the order of the variables *country* and *survived* above – i.e. `> plot(table(survived, country))` we would obtain the chart shown opposite, which is not as informative as the one above.

It is frequently the case with two charts of the same data can differ in how useful they are. It is advisable in statistics to generate multiple charts – the first one created may not be the most informative.

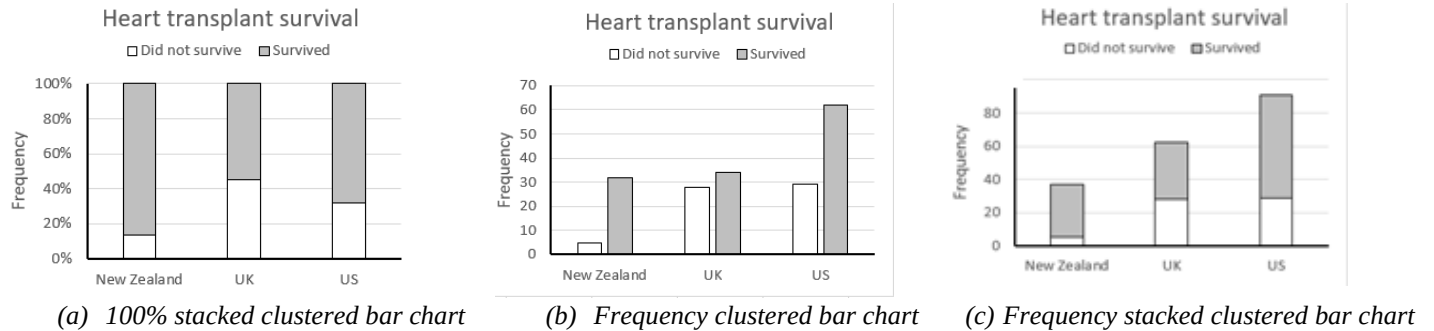


Note that the width of the “no” and “yes” stacks above are proportional to the sample sizes. Thus, we can see that the yes:no ratio is about 2:1

Clustered bar charts

The type of chart constructed here are example of clustered bar charts and these are useful for visualising two categorical variables. Specifically, the two examples above are 100% stacked bar charts. Stacked, because the bars representing the levels of one variable are stacked rather than side by side and 100%, because each stack is the same height, which facilitates the comparison of percentages rather than frequencies.

Other variations of the clustered bar chart as well as the 100% stacked clustered bar chart include the frequency clustered bar chart, and the stacked clustered bar charts – examples of each (created in Excel) are shown below for the current problem.



(b) is better than (a) at showing the relative size of the samples (e.g. we can see there are more US subjects than New Zealand subjects) but it is not as good as (a) at showing how survival rates differ across countries. (c) could be seen as a compromise between (a) and (b).

Exercise

See if you can generate charts similar to these in R.

Does survival rate differ by country?

H_0 : Survival rate is the same in all countries

H_A : Survival rate is not the same in all countries (at least two countries differ)

We test this hypothesis using a χ^2 test of independence.

```
> chisq.test(survived, country)

Pearson's Chi-squared test

data: survived and country
X-squared = 10.604, df = 2, p-value = 0.004983
```

As $p < 0.05$ we reject the null hypothesis of independence

I.e. the survived : did not survive ratio is not the same for the three counties

The survival percentages in the three countries are New Zealand (86.4%), UK (54.8%) and US (68.1%)

For the purposes of this module, we will not consider any further analysis to determine which countries differ from which. We will speculate that all three differ.

Note that the χ^2 test of independence is a non-parametric test. It applies to two nominal / binary variables. Such variables are never normally distributed so checks on normality are not applicable.

Example 2

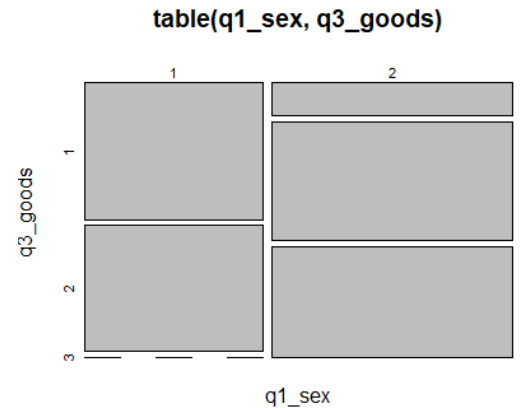
In our customer satisfaction survey data, do males and females differ in their choice of goods?

Remember our coding (1. Male 2. Female and 1. Electronics 2. Jewellery 3. Fragrances)

A cross-tab and plot of the data is generated

```
> table(q1_sex, q3_goods)
      q3_goods
q1_sex  1  2  3
  1  12 11  0
  2   4 14 13
> plot(table(q1_sex, q3_goods))
```

There is a strong indication that men and women differ in their goods preference. Note that men buy no fragrances and few women buy electronics.



```
> chisq.test(q1_sex, q3_goods)

Pearson's Chi-squared test

data:  q1_sex and q3_goods
X-squared = 16.538, df = 2, p-value = 0.0002564
```

The χ^2 test of independence confirms that the variables are not independent

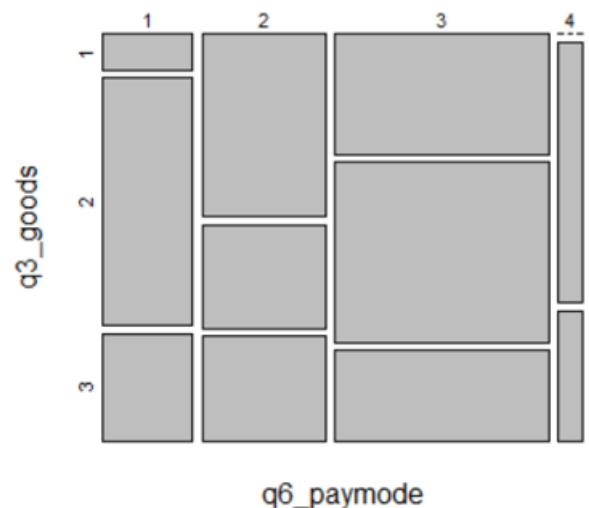
Example 3

Do the methods customers use to pay for goods (credit card, PayPal, other) differ by goods types?

Remember our coding

(1. Credit card 2. Debit card 3. PayPal 4. Other)

```
> table(q6_paymode, q3_goods)
      q3_goods
q6_paymode 1  2  3
  1   1  7  3
  2   7  4  4
  3   8 12  6
  4   0  2  1
> plot(table(q6_paymode, q3_goods))
```



Remember, the extent to which the blocks do not align horizontally is indicative that the variables are not independent – so it appears as the two variables might be independent here.

Note however, the narrowness of the fourth level of payMode (other) – this reflects the small numbers in the “other” category (3 respondents).

A χ^2 test of independence does not find that the variables are independent ($p = 0.3776 > 0.05$)

```
> chisq.test(q6_paymode, q3_goods)

Pearson's Chi-squared test
data: q6_paymode and q3_goods
X-squared = 6.4217, df = 6, p-value = 0.3776

Warning message:
In chisq.test(q6_paymode, q3_goods) :
  Chi-squared approximation may be incorrect
```

In addition, note that there is a warning about the approximation possibly being incorrect. This sometimes happens if the number of cells (entries in the table above - or equivalently, the number of blocks in the chart) is large compared to the amount of data.

In this example we have 12 entries in the table and only 57 respondents. Of course, as we have not asserted that the variables are independent (we do not reject the null hypothesis) we need not be concerned with the warning here. But it is possible that we might have established a lack of independence between the two variables had we not had so many cells.

Sometimes, we may be able to reduce the size of the table in some logical way. An obvious choice here might be to omit the “other” category.

Constructing a table and plot is left as an exercise. The χ^2 test of independence output is shown below.

```
> chisq.test(q6_paymode[q6_paymode<4], q3_goods[q6_paymode<4])

Pearson's Chi-squared test
data: q6_paymode[q6_paymode < 4] and q3_goods[q6_paymode < 4]
X-squared = 4.9945, df = 4, p-value = 0.2879

Warning message:
In chisq.test(q6_paymode[q6_paymode < 4], q3_goods[q6_paymode < 4] :
  Chi-squared approximation may be incorrect
```

So, omitting the “other” category did not lead to a finding that the variables are not independent.

Exercise

Decide if the type of goods bought by customers paying using cards (credit or debit) is different to the types of goods bought by customers paying using PayPal.

N.B. You do need some kind of rationale to justify combining categories. You certainly can make an argument for discounting others, or for combining credit and debit cards together. But it would be difficult to justify say combining say PayPal and credit cards in one group and debit cards in another – even if this led to nice groupings of the data.

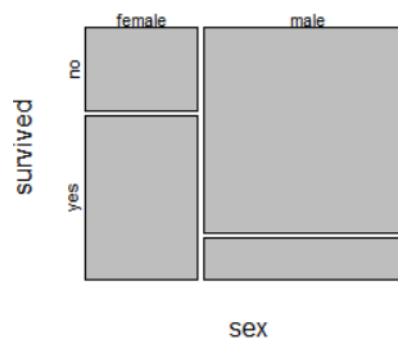
As another example, suppose exam results (pass/fail) were to be analysed for a group of students of different disciplines (computing, engineering, business and law) you may elect to combine the computing and engineering students and/or the business and law students if the samples were too small. Of course, there is a subjective element to making such decisions on how to collapse groups

Example 3

Did the percentage of male passengers who survived the Titanic sinking differ from the percentage of females.

The Titanic data is [available](#) as a CSV file.

```
> table(sex, survived)
      survived
sex      no yes
female 156 307
male   708 142
> plot(table(sex, survived))
```



The plot conveys a number of facets about the data.

- There were substantially more males than female passengers.
- There were more females than males who did survive.

From the table above we can calculate the percentage of male survivors to be $334 / (1409 + 334) = 0.19$
 And the percentage of female survivors to be $327 / (158 + 327) = 0.67$

So about 1 in 5 males survive whereas the number of females who survive is about 2 in 3

To decide if there is a statistically significant difference between the two ratios we carry out a test of independence of the variables survived and sex is shown below – the results of which are shown below.

(Arguably the passengers on the Titanic constitute a population rather than a sample – in which case statistical inference does not apply)

```
> chisq.test(survived, sex)

Pearson's Chi-squared test with Yates' continuity correction

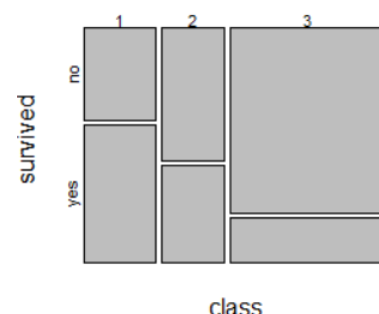
data:  survived and sex
X-squared = 421.2, df = 1, p-value < 2.2e-16
```

As the p value < 0.05 the variables survived and sex are not independent – the percentage of females who survived is not the same as the percentage of males who survived.

Example 3

Is there evidence that passenger class on the Titanic was a factor in survival propensity?

```
> table(class, survived)
      survived
class  no yes
  1    129 193
  2    161 119
  3    574 137
> plot(table(class, survived))
```



The % of 1st, 2nd and 3rd class survivors is 60%, 43% and 19% respectively and these percentages are statistically significantly different. (*Exercise*)

Note that the analysis used in this example makes no use of the ordinal structure of the variable survived. Usually, we would try and use all available information in our analysis as a sub-optimal test may not yield a statistically significant finding. Fortunately, this was not the case in this example.